CO2 - AT 3 - class TEST

G.chandra sekhar Reddy
192472173
ITA0502 CV

1) Image Enhancement in the Spatial Domain

Definition: Image enhancement in the spatial domain refers to improving the visual quality of an image by directly modifying the pixel values. The main objective is to make the image more suitable for human observation or further image processing.

Gray-Level Transformations

Gray-level transformations modify the intensity values of pixels.

Types:-

Image Negative:- Reverses brightness values, useful in medical imaging.

Log Transformation:- Enhances dark regions.

Power-law Transformation:- Adjusts brightness and contrast.

Contrast Stretching:- Expands intensity range to improve visibility.

Advantages:-

* Simple and fast
* Improves brightness and contrast.

2) Image smoothing vs Image sharpening.

Image smoothing

Image smoothing reduces noise by averaging neighboring pixels.

working principle A smoothing filter replaces the center pixel with the average or weighted average of surrounding pixels.

Types :-

- mean filter
- Gaussian filter
- median filter.

Advantages

- Removes random noise.
- Produces smooth images.

Limitations

- Blurs edges.
- Reduces fine details

Image sharpeing:-

Types: • Laplacian filter
- Sobel filter
- Prewitt Filter

Advantages:-
- Enhances edges.
- Improves feature visibility

Limitations;- may amplify noise.

(3) Image Enhancement in Frequency Domain

Definition

Frequency domain enhacement processes images after converting them into the frequency domain using the Fourier Transform.

Steps:-

1. Convert image using Fourier Transform.
2. Apply frequency filter.
3. Perform Inverse Fourier Transform.
4. Obtain enhanced image.

Low - Pass Filter (LPF)

working:-

- Passes low-frequency components.
- Blocks high-frequency components.

Effects:-

- Smooths the image.
- Removes noise.

Applications

- fingerprint recognition.
- Industrial inspection
- OCR

Advantages:-

- Efficient for large images.
- provides better control over enhancement.

(4) Image segmentation

Definition

Image segmentation divides an image into meaningful regions for easier analysis.

Techniques

1. Point Detection

Detects isolated pixels with different intensity.

Applications:

- Fault detection.
- medical imaging.

2. Line Detection

Detects horizontal, vertical, and diagonal lines.

Applications

- Road detection.
- Text recognition.

3. Edge Detection

Detects boundaries using intensity changes.

methods

- Sobel
- Prewitt
- Canny

Applications

- object recognition
- face detection.

5) Edge detection, Edge Linking and Boundary Detection

Edge Detection:

Edge detection identifies sudden changes in image intensity

methods

- Sobel
- Prewitt
- Roberts
- Canny

Importance

- Detects object boundaries.
- Reduces unnecessary information.

Edge Linking.

Edge linking connects broken edge pixels into continuous edges.

methods

- Hough Transform
- Graph-based methods

Importance

- Produces complete object outlines,
- Improves segmentation.